



Indywidualny Plan Badawczy (IPB) Individual Research Plan (IRP)

1. Podstawowe dane / Basic data

Podstawowe informacje dotyczące doktoranta i realizowanej rozprawy doktorskiej. W przypadku promotora spoza Politechniki Wrocławskiej należy podać afiliację. / *Basic information on the doctoral student and the doctoral dissertation underway. In the case of a supervisor from outside Wrocław University of Science and Technology, affiliation should be provided.*

Imię i nazwisko / Name, surname	Grzegorz Piotrowski
Dyscyplina kształcenia / Discipline of education	informatyka techniczna i telekomunikacja / information and communication technology
Wydział / Faculty	W4 Wydział Informatyki i Telekomunikacji / Faculty of Information and Communication Technology
Katedra / Department	K46 Katedra Sztucznej Inteligencji / Department of Artificial Intelligence
Data rozpoczęcia kształcenia / Date of commencement of the doctoral education	01.10.2025
Nr ORCID	0009-0004-6013-0461
Promotor/ promotorzy / Supervisor(s)	dr hab. inż. Tomasz Jan Kajdanowicz, prof. uczelni, Politechnika Wrocławska (K46, W4), ORCID 0000-0002-8417-1012
Promotor pomocniczy / Assistant supervisor	—

2. Temat rozprawy doktorskiej / Topic of doctoral dissertation

Propozycja tematu rozprawy doktorskiej. Przedstawiony temat nie musi stanowić ostatecznego tytułu, jaki zostanie nadany rozprawie doktorskiej złożonej na zakończenie kształcenia w Szkole Doktorskiej. / *Proposed topic for a doctoral dissertation. The presented topic does not have to constitute the final title of the doctoral dissertation submitted at the end of education at the Doctoral School.*

PL: Reprezentacje wewnętrzne jako miara i predyktor przenoszenia do świata rzeczywistego polityk agentów ucieleśnionych uczonych w cyfrowych bliźniakach z neuronowej rekonstrukcji scen, przy ograniczonym budżecie danych rzeczywistych

EN: Internal representations as a measure and predictor of the real-world transfer of embodied policies learned in neural-reconstruction digital twins under a limited real-data budget

3. Harmonogram przygotowania rozprawy doktorskiej / Schedule for the preparation of the doctoral dissertation

Harmonogram przygotowania rozprawy doktorskiej z wyszczególnieniem etapów i miejsc prowadzonych badań, w tym terminów zakończenia badań częściowych (semestr kształcenia) i opracowania ich wyników oraz opcjonalnie w zależności od rodzaju prowadzonych badań, wybranych między innymi spośród następujących:



- realizacja badań (w tym przygotowanie stanowiska badawczego, przygotowanie materiałów do badań, wykonanie pomiarów/eksperymentów, opracowanie otrzymanych wyników),
- przygotowanie i wysłanie artykułu naukowego do redakcji czasopisma naukowego,
- wysłanie zgłoszenia patentowego do właściwego urzędu,
- opracowanie propozycji wdrożenia dla przemysłu,
- udział w konferencjach, warsztatach lub szkołach letnich/zimowych i prezentacja wyników badań,
- przygotowanie wniosku i wystąpienie o grant,
- mobilność krajowa lub zagraniczna doktoranta (w tym konsultacje naukowe i wizyty studyjne w ośrodkach krajowych lub zagranicznych).

A schedule for the preparation of the doctoral dissertation detailing the stages and locations of the research to be conducted, including the dates for completing the sub-studies and developing their results, and optionally, depending on the type of research to be conducted, selected among the following:

- *execution of the research (including preparation of the research site, preparation of materials for the research, execution of measurements/experiments, elaboration of the obtained results),*
- *preparation and submission of a scientific article to the editor of a scientific journal,*
- *submission of a patent application to the relevant office,*
- *development of an implementation proposal for the industry,*
- *participation in conferences, workshops or summer/winter schools and presentation of research results,*
- *preparation of an application and application for a scholarship,*
- *domestic or foreign mobility of the doctoral student (including scientific consultations and study visits to domestic or foreign centres).*

Okres realizacji zadania (semestr kształcenia) / Task implementation period (semester)	Zwięzły opis zadania / Brief description of the task
Semestr 1	<i>(Oct 2025 – Feb 2026)</i> Research target: define the problem and map the state of the art. T1.1 Systematic literature review in four areas: (a) sim-to-real transfer and domain randomization, (b) learning under distribution shift and domain adaptation, (c) neural scene reconstruction (NeRF, 3D Gaussian Splatting), (d) learning-based visual navigation. <i>D:</i> annotated bibliography and a written research-gap statement (basis of §6). T1.2 Identification of the open problem: how much real data sim-to-real transfer needs, and whether reconstructed scenes can replace real data. <i>D:</i> problem statement and preliminary research questions, discussed with the supervisor. T1.3 Doctoral School courses of semester 1. <i>D:</i> completed courses. Place: K46.



Semestr 2	<i>(Mar – Sep 2026)</i> Research target: formulate the research plan and select the experimental basis. T2.1 Formulation of the representation-level thesis, research questions RQ1–RQ4 (measure, predict, use, generalize) and hypotheses H1–H4 with measurable decision rules and evaluation tiers A–C. <i>D:</i> §7 and §9 of this IPB. T2.2 Comparative survey of public benchmarks and datasets (e.g. ScanNet++, HM3D), simulators (e.g. Habitat-Sim) and 3D Gaussian Splatting implementations against the needs of RQ1–RQ4 (real video + reference scan, licence, compute cost). <i>D:</i> tool-selection note with a shortlist for Stage I. T2.3 Individual Research Plan agreed with the supervisor. <i>D:</i> this IPB, signed and submitted by 30.09.2026. Place: K46.
Semestr 3	<i>(Oct 2026 – Feb 2027)</i> Stage I – paired-frame benchmark (RQ1, H1). T3.1 Final choice of ≥ 10 public scenes with real captures and reference scans, simulator and existing 3DGS pipeline from the T2.2 shortlist; twins built at ≥ 4 capture budgets; paired real/twin frames of the same pose with fidelity and cost logs; pre-registration of the evaluation protocol (§7). <i>D:</i> benchmark v0 (data scripts, code) and protocol in the repository. Place: K46, WCSS. T3.2 Validation stand: written agreement on robot access (planned cooperation with the K29 Denali laboratory, or another PWr platform) and sensor decision (SzD Minigrant application if a sensor must be bought). <i>D:</i> agreement, by Nov 2026. T3.3 Layer-wise gap measurement (linear CKA, linear probes) for ≥ 10 frozen encoders; pilot of policy-zoo training that fixes the zoo size. <i>D:</i> first H1 results (concentration, budget trend, encoder differences) and pilot report, by Feb 2027.
Semestr 4	<i>(Mar – Sep 2027)</i> Stage I completed, Stage II started (RQ1–RQ2, H1–H2). T4.1 Navigation policies on each frozen encoder trained in the twins and tested in proxy reality (tier A); robustness ranking vs. twin→real success (first result for P1 by Apr 2027); gap localization inside twin-trained policies; generation of the policy zoo (≥ 200 navigation policies with twin and proxy-real labels). <i>D:</i> H1 completed on tier A; zoo v1, by Sep 2027. Place: K46, WCSS. T4.2 Prepare and submit article P1 to NeurIPS 2027 (paired-frame benchmark and gap localization in frozen encoders, H1(a–c), with the first H1(d) result; RQ1; Datasets & Benchmarks or main track). <i>D:</i> submission, May 2027 (§11); benchmark released with the paper. T4.3 Grants and international activity: NCN PRELUDIUM application; poster at an international summer school; foreign co-author sought for P2; NAWA Bekker application for the sem. 6 visit. <i>D:</i> applications, poster.



Semestr 5	(Oct 2027 – Feb 2028) Mid-term evaluation (Nov 2027): autoreferat (≤ 5 pages, ~mid-Oct 2027) and 15-min presentation. Stage II – transfer forecasting (RQ2, H2). T5.1 Transfer predictors on the zoo (graph neural network over layers, weight-space metanetwork) vs. baselines (twin SR, PSNR/LPIPS, simulator-level predictivity) on held-out scenes; hidden-state failure monitors calibrated conformally in the twin and tested in proxy reality (tier A); Stage III set-up (allocation levers implemented, Jan–Feb 2028). <i>D:</i> H2 completed. Place: K46, WCSS. T5.2 First robot validation campaign (tier C, 8 zoo policies, 2 PWR environments): do forecasts rank real outcomes as on tier A? <i>D:</i> validation report. Place: Denali. Without robot access, the campaign moves to sem. 7; H1–H2 are decided on tier A either way. T5.3 Article P2 (forecasting transfer and failure from internals) submitted to CVPR 2028 (~Nov 2027; ICLR 2028, ~late Sep 2027, if ready earlier). If P1 was rejected, its revision goes to CVPR 2028. <i>D:</i> submission(s).
Semestr 6	(Mar – Sep 2028) Stages III–IV – using the forecasts; cross-task transfer (RQ3–RQ4, H3–H4). T6.1 (a) <i>Budget allocation:</i> forecast-guided weighting of twin data, capture selection and rollout selection vs. uniform and random allocation; budget–performance curves (capture minutes + real rollouts). (b) <i>Manipulation:</i> public policies with published SIMPLER paired sim/real results (≥ 12 policy–task pairs); the navigation predictor applied without retraining, compared with generic-simulator SRCC and a frozen world-model evaluator. <i>D:</i> H3 completed (tier A, by Apr 2028), H4 completed (tier B, by Sep 2028). Place: K46, WCSS. T6.2 Foreign research visit (3 months, NAWA Bekker; if not funded, a 5–30-day Erasmus+ short-term mobility to a European lab; hosts in §12): joint experiments. <i>D:</i> visit report. T6.3 Article P3 (budget allocation with the first cross-task results) submitted to NeurIPS 2028 (~May 2028); if not ready, to ICLR 2029 (~late Sep 2028). <i>D:</i> submission.
Semestr 7	(Oct 2028 – Feb 2029) Consolidation (RQ1–RQ4). T7.1 Second robot validation campaign (tier C) of the forecasts and of guided rollout selection; final analysis of H1–H4 across tiers. <i>D:</i> final results. Place: K46, Denali. T7.2 Release of code, paired-frame benchmark and policy zoo. <i>D:</i> public repository. T7.3 Optional article P4 (consolidated study) to ICLR 2029 (~late Sep 2028) or CVPR 2029 (~Nov 2028); start of dissertation writing. <i>D:</i> submission (optional), dissertation outline.
Semestr 8	(Mar – Sep 2029) T8.1 Editing of the doctoral dissertation; final version. <i>D:</i> dissertation submitted by 30.09.2029 (§4). T8.2 Responses to reviews and camera-ready versions of pending articles. <i>D:</i> revised manuscripts.

4. Termin złożenia rozprawy doktorskiej / Date of the doctoral dissertation submission



Planowany termin złożenia rozprawy doktorskiej nie może przekraczać daty zakończenia ósmego semestru kształcenia w Szkole Doktorskiej. / A planned date for submission of the dissertation, which must not exceed the date of completion of the eighth semester of education in the Doctoral School.

30.09.2029

(rekomenrowana data to ostatni dzień z okresu 4 lata kształcenia w Szkole Doktorskiej, np. jeżeli kształcenie rozpoczęło się 1.10.2025, to proszę wpisać 30.09.2029 /

The recommended date, which marks the final day of the four-year doctoral training period. For example, if your training began on October 1st, 2025, please enter September 30th, 2029)

5. Uzasadnienie wyboru tematu rozprawy doktorskiej / Justification for choosing the topic of the doctoral dissertation

Opis zawierający uzasadnienie wyboru tematu rozprawy doktorskiej uwzględniający potencjalne obszary zastosowania planowanych wyników rozprawy doktorskiej (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description containing the justification for the selection of the topic of the doctoral dissertation, taking into account the potential areas of application of the planned results of the doctoral dissertation (at most 1 page, font 11, line spacing 1).

Modern machine learning is limited less by models than by **data in the target domain**. Embodied agents (mobile robots, manipulators) learn strong behaviour only from very large amounts of experience. In the real world such data is slow, expensive and sometimes risky to collect, so a policy is usually trained in simulation and then deployed on real sensor data, where it often fails (the **sim-to-real gap**).

Digital twins built by neural scene reconstruction (3D Gaussian Splatting, Neural Radiance Fields) turn a few minutes of real video into a photorealistic, interactive model of the target scene, in which a policy can be trained at scale. Building such twins has quickly become routine: dozens of papers a year and more than ten open-source twin simulators. The open question has moved from *how to build a twin* to *when a policy trained in it can be trusted in reality*. Today this is answered in two unsatisfying ways: by running the policy in the real world (slow and costly, which is what the twin was meant to avoid) or by image-quality scores of the reconstruction, which need not track real performance (lower-fidelity simulation can even transfer better). Neither says **where** the gap arises, and neither tells how to spend the few real observations one can afford.

Why representations. Whatever a policy does, it does through its internal representations: the features its encoder and layers compute from an image. This is where reconstructed and real inputs either meet or diverge. Machine learning now has mature tools to read representations: layer-wise probing, representational similarity (e.g. centred kernel alignment), and learning on network weights (metanetworks over model populations). They have been used to predict the accuracy and generalization of image classifiers and other networks from their weights, but not the real-world transfer of policies trained in twins. The dissertation asks whether this transfer can be **measured, localized and predicted from representations** without real rollouts, and whether such forecasts let a small real-data budget be spent where it matters: which twin data to weight, what to capture, and which real trials to run.

Why this topic, and why in this discipline. The object is a learning and evaluation methodology, not a robot or a simulator, which places it in *information and communication technology* (representation learning, learning under distribution shift, model analysis). The student uses existing twin pipelines and simulators and does not develop new ones. **Visual navigation** is the primary testbed, because public scene datasets with real captures allow controlled, reproducible measurement without an own robot; **robotic manipulation** tests that the findings carry over to another task. Trials on a real robot (planned cooperation with a PWR robotics laboratory) validate, but do not decide, the conclusions. The topic fits the Department of Artificial Intelligence (K46), whose research groups include representation learning on graphs and models in weight space. The computation runs on the GPU



infrastructure available to PWR researchers (WCSS, PLGrid). A benchmark of paired real and reconstructed data and a measured answer to "which representations survive reconstruction" are useful whatever the hypotheses' outcome.

Potential application areas: faster and cheaper deployment of learned robots in new buildings and workcells (service, assistive, warehouse and inspection robotics, flexible manipulation); pre-deployment screening of candidate policies and safety monitors that flag likely failures; choosing which visual foundation models to use in twin-based training; and, beyond robotics, any system trained on reconstructed or synthetic data and deployed on real sensor data (industrial digital twins, perception for autonomous systems).

6. Zarys aktualnego stanu badań w tematyce rozprawy doktorskiej / Outline of the current state of research in the topic of the doctoral dissertation

Przegląd literaturowy zawierający opis aktualnego stanu badań w tematyce rozprawy doktorskiej uwzględniający odwołania do najważniejszych pozycji literaturowych związanych z tematyką rozprawy doktorskiej (maks. 2 strony, czcionka 11 pt., interlinia 1). / A literature review containing a description of the current state of research in the subject of the doctoral dissertation, including references to the most important literature items related to the subject of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

Digital twins have become a commodity. Photorealistic simulators such as Habitat [1] and domain randomization [2] made large-scale training of embodied agents possible. 3D Gaussian Splatting (3DGS) [3] now turns a short phone or camera capture into a real-time, photorealistic *digital twin*. Twin-trained policies transfer to real robots in navigation (EmbodiedSplat [4], Vid2Sim [5], GaussGym [6]) and manipulation (RialTo [7], SplatSim [8]), and GPU simulators ship twin environments (ManiSkill3 [9]). Building twins is therefore no longer the bottleneck. The open question is **when a twin-trained policy will work in reality, and why**.

Predicting real performance from simulation. Kadian et al. [10] introduced the Sim-vs-Real Correlation Coefficient (SRCC) and showed that simulation can poorly predict real navigation results. Truong et al. [11] found that lower-fidelity simulation can transfer better, so image fidelity is not the same as transfer. SIMPLER [12] reports strong sim-real correlation for manipulation evaluation, and pretrained visual representations rank similarly in simulation and reality [13]. All of these measure transfer at the level of *simulators* and success rates, after deployment.

The representation gap. Sim-and-real co-training works with few real demonstrations [14], domain adaptation aligns sim and real features [15], and a mechanistic analysis finds that "structured representation alignment" is the primary effect of co-training [16]. ScanNet++ [17] pairs real captures of rooms with reference scans. Linear probes [18] and centered kernel alignment (CKA) [19] measure what each layer encodes and how similar two representations are. Frozen image and video encoders (DINOv2 [20], V-JEPA 2 [21]) are common policy backbones, and probing of vision-language-action models shows that action fine-tuning degrades their visual representations [22]. However, **we found no study that measured, on paired real and twin-rendered frames of the same pose, where inside encoders and policies the twin-vs-real gap arises, or how it shrinks with the capture budget**.

Predicting performance from model internals. Accuracy under distribution shift can be estimated from unlabeled target data [23]. A network's accuracy can be predicted from its weights alone [24], and populations of trained models ("model zoos" [25]) are now inputs to weight-space learning: equivariant metanetworks [26] learn directly on weights, and graph neural networks over the computational graph [27] predict generalization. They have been tested on image classifiers and implicit neural representations, **not on embodied policies or twin-to-real transfer**.

Failure prediction and calibrated monitoring. Conformal prediction [28] gives distribution-free guarantees. FAIL-Detect [29] detects failures of imitation policies without failure data and calibrates thresholds with conformal prediction, and SAFE [30] finds that the features of vision-language-action



models encode task success and failure across tasks. These monitors are calibrated and tested in the *same* domain. Whether a monitor calibrated in the twin keeps its coverage on real data has not been studied.

Spending a small real-data budget. Data attribution estimates how training examples affect predictions [31], and CUPID [32] ranks robot demonstrations by their influence on closed-loop success. Prediction-powered inference [33] combines many cheap predictions with a few labels, and SureSim [34] uses it to correct simulated evaluation with a few paired real trials. Learned world models are also proposed as policy evaluators [35]. These tools treat the policy as a black box; none uses its internal representations to decide which twin data to weight, what to capture or which real rollouts to collect.

Research gap. Twins are widely available, but there is **no representation-level account of how twin-trained policies transfer to reality**. Missing are (i) a paired real/twin benchmark that localizes the gap inside encoders and policies across capture budgets, (ii) predictors of real transfer and failure from policy internals (weights, hidden states) that beat simulator-level and image-fidelity baselines, (iii) methods that use such forecasts to spend a limited real-data budget, and (iv) evidence that these measures carry over from navigation to manipulation. This dissertation addresses that gap.

References

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7. Pytania i hipotezy badawcze / Research questions and hypotheses

Opis najważniejszych pytań i hipotez badawczych związanych z realizowaną rozprawą doktorską (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description of the most important research questions and hypotheses related to the doctoral dissertation (at most 1 page, font 11, line spacing 1).

Thesis. How well embodied policies trained in neural-reconstruction digital twins transfer to the real world can be **measured, localized and predicted from their internal representations**, and this lets a limited real-data budget be spent where it matters. Navigation is the primary testbed; manipulation is the cross-task test. RQ1–RQ2 (H1–H2) are the core of the dissertation; RQ3–RQ4 (H3–H4) build on them.

Setting. A *twin* is a 3D Gaussian Splatting reconstruction of a real scene built from a short capture; the *capture budget* is the number of capture minutes (views). Evaluation tiers: **(A)** proxy reality on public scene datasets with real captures and reference scans (≥ 10 scenes, some held out), which decides the hypotheses; **(B)** public real-world datasets and published real evaluations with real outcomes; **(C)** real-robot trials, validation only. A *policy zoo* (≥ 200 navigation policies varying scene, capture budget, architecture and training data) supplies the population for RQ2–RQ4. Tests are one-sided, $\alpha = 0.05$, bootstrapped over scenes, Holm-corrected within each hypothesis and pre-registered in the repository. All thresholds are design choices fixed in that pre-registration.

RQ1 (measure and localize). Where inside frozen encoders and twin-trained policies does the twin-vs-real gap arise, and how does it depend on the capture budget?

H1. On paired real and twin-rendered frames of the same pose, the representation gap (1 – linear CKA, and the drop of linear-probe accuracy) (a) is *concentrated*: the third of layers with the largest gap holds $\geq 50\%$ of the summed gap in $\geq 80\%$ of scenes; (b) decreases monotonically with the capture budget (trend test over ≥ 4 budgets); (c) differs between ≥ 10 frozen encoders (Friedman test); and (d) the encoders' robustness ranking predicts their downstream twin $\rightarrow$ real success rate (SR) with Spearman $\rho \geq 0.6$.

RQ2 (predict). Can a twin-trained policy's real-world transfer and failures be forecast from its internals (hidden states, weights) without real rollouts?

H2. (a) A predictor trained on the zoo (GNN over layers or weight-space metanetwork) predicts the twin $\rightarrow$ real SR gap on held-out scenes with $\geq 20\%$ lower mean absolute error than the best of three baselines (twin SR alone, image fidelity PSNR/LPIPS, simulator-level predictivity); the lower 95% bound of the relative reduction must be > 0 . (b) Hidden-state failure monitors calibrated conformally in the twin at 90% coverage keep coverage $\geq 85\%$ on real data ($\epsilon = 5$ pp).

RQ3 (use). Can these measures and forecasts allocate a limited real-data budget: which twin data to weight, what to capture and which real rollouts to collect?

H3. Forecast-guided weighting, capture and rollout selection reaches the target real SR τ with $\geq 30\%$ less real data than uniform or random allocation (capture minutes + real rollouts, converted to one operator-time cost at a pre-registered rate): $B_{\text{guided}}(\tau) \leq 0.7 \cdot B_{\text{uniform}}(\tau)$ on fitted budget–performance curves, with the upper 95% bound of the ratio < 1 . τ is fixed in the pre-registration as the SR that uniform allocation reaches at the largest budget.

RQ4 (generalize). Do the gap measures and predictors transfer across tasks and simulator families?

H4. (a) A predictor trained on the navigation zoo, applied without retraining to manipulation policies with published paired sim/real results (≥ 12 policy–task pairs), ranks them by real outcome with Spearman $\rho \geq 0.5$. (b) Its rank correlation with real outcomes exceeds that of a generic simulator (Sim-vs-Real Correlation Coefficient, SRCC) and of a learned world-model evaluator (95% CI of each paired difference excludes 0).



Decision rule. A hypothesis is supported when all its parts pass on tier A; tiers B and C report agreement and are not used to tune thresholds.

8. Wkład spodziewanych wyników dla rozwoju dyscypliny naukowej / Contribution of the expected results to development of scientific discipline

Krótki opis spodziewanego wkładu wyników rozprawy doktorskiej dla rozwoju dyscypliny naukowej, w której realizowana jest rozprawa doktorska (maks. 1 strona, czcionka 11 pt., interlinia 1). / Brief description of the expected contribution of the results of the doctoral dissertation to the development of the scientific discipline in which the doctoral dissertation is carried out (at most 1 page, font 11, line spacing 1).

The dissertation is carried out in **information and communication technology** (*informatyka techniczna i telekomunikacja*), in the area of machine learning. Its object is the **internal representations** of encoders and embodied policies trained in digital twins built by neural reconstruction of real scenes, used to measure, localize and predict how well the policies transfer to reality. Visual navigation is the primary testbed and robotic manipulation the cross-task test. Building twins or simulators, control design and mechanical engineering are not the object of the research.

Key original contribution (core, RQ1–RQ2, H1–H2). To our knowledge, the first systematic test of whether the twin-to-real transfer of embodied policies can be **read from their representations**: where in the network the gap arises, how it shrinks with the capture budget, and how large it will be for a given policy, forecast from its hidden states or weights without real rollouts.

Contributions to machine learning methodology:

1. **Measuring and localizing the reconstruction gap (core; RQ1, H1).** A layer-wise analysis of paired real and twin-rendered observations (representational similarity, probing) that locates the gap inside frozen visual encoders and policies, relates it to the capture budget, and ranks encoders by robustness to reconstruction artifacts, tested against downstream transfer.
2. **Forecasting transfer and failure from internals (core; RQ2, H2).** Predictors learned over a population of twin-trained policies (graph networks over layers, weight-space metanetworks) that estimate a policy's real performance before deployment, and hidden-state failure monitors whose statistical coverage is checked across the twin-to-real shift. This brings weight-space learning and model analysis to embodied policies and distribution shift.
3. **Representation-guided use of a real-data budget (RQ3, H3).** Methods that use the measures and forecasts to weight twin training data, choose what to capture and select which real rollouts to collect, quantified by budget–performance curves against uniform and random allocation.
4. **Generalization of the measures (RQ4, H4).** Evidence of whether representation-based predictors carry over from navigation to manipulation and across simulator families, compared with simulation-based predictivity (SRCC) and a learned world-model evaluator.

Contributions in engineering terms (ITiT):

5. **Benchmark, data and software (supports all RQs).** A public benchmark of paired real and twin-rendered frames at graded capture budgets, a released zoo of twin-trained policies with twin and real outcomes, and evaluation code with pre-registered protocols, built on existing open-source twin pipelines and simulators. It gives other groups a reproducible way to test sim-to-real claims and to screen policies and encoders before a costly real deployment.

Significance for the discipline. Real target-domain data is the main cost of deploying learned systems, and trained models are increasingly deployed on data unlike their training data. Testing whether transfer can be diagnosed and forecast from a model's internals, and whether such forecasts save real data, is relevant to machine learning under distribution shift and to model evaluation in general, beyond robotics.



Dissemination. Results are planned for peer-reviewed conferences worth 200 points and assigned to ITiT on the ministerial list of 5.01.2024: NeurIPS and ICLR (machine learning) and CVPR (computer vision), with ICML, ICCV and ECCV for resubmissions, as in §3, §11 and §12. P1 (RQ1, benchmark and localization) targets NeurIPS 2027; P2 (RQ2) ICLR or CVPR 2028; P3 (RQ3–RQ4, including manipulation) NeurIPS 2028.

9. Planowane metody badawcze / Planned research methods

Krótki opis metod badawczych, które są będą użyte w czasie realizacji rozprawy doktorskiej dla (maks. 2 strony, czcionka 11 pt., interlinia 1). / Brief description of the research methods that will be used during the implementation of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

The research follows four stages (I–IV) that match §3 and the research questions of §7. For each stage we give the method, data, metric and success criterion. The student uses existing open-source 3D Gaussian Splatting (3DGS) pipelines and simulators and does not develop new ones. Stages I–II (RQ1–RQ2) are the core; Stages III–IV (RQ3–RQ4) build on them.

Data and evaluation tiers. *Tier A (proxy reality; decides the hypotheses):* ≥ 10 public indoor scenes that have real captures and a reference laser scan, e.g. ScanNet++ [17]. The reference scan, rendered in a GPU simulator (Habitat [1] or an Isaac-based 3DGS renderer such as GaussGym [6], chosen in T3.1), is the "real" target domain. The twin is a 3DGS reconstruction built from a subsample of the scene's real capture. The **capture budget** (≥ 4 levels of capture minutes or views) is set by subsampling. Some scenes are held out. Frame-level measures use the real images; closed-loop outcomes use the rendered scan, and the gap between scan renders and real images is measured with the same tools, so the proxy's own error is known. *Tier B (real outcomes):* public real-world navigation datasets for representation metrics and published paired sim/real evaluations of manipulation policies (SIMPLER [12]). *Tier C (validation only):* ≥ 2 PWR environments with a mobile robot (planned cooperation with the K29 "Denali" Autonomous Robots Laboratory; agreement in T3.2), RGB-D camera and wheel odometry. *Compute:* WCSS Lem (NVIDIA H100) and PLGrid allocations (to be applied for).

Stage I – paired-frame benchmark and gap localization (RQ1, H1; sem. 3–4).

- *Benchmark:* for every scene and capture budget we render twin frames at the camera poses of held-out real frames, which gives **paired real/twin frames of the same pose**, plus per-scene fidelity (PSNR, SSIM, LPIPS, depth error) and cost (capture minutes, GPU-hours).
- *Encoders:* ≥ 10 frozen image and video encoders used as policy backbones (e.g. DINOv2 [20], CLIP-type, V-JEPA 2 [21]), plus the encoders inside twin-trained policies.
- *Method:* for each layer, **linear CKA** [19] between real and twin activations, and **linear probes** [18] trained on twin features and tested on real features (targets: depth, semantic class, relative goal direction). Gap = $1 - \text{CKA}$ and the probe-accuracy drop.
- *Success criterion (H1):* concentration of the gap in layers, monotonic decrease with the capture budget, differences between encoders, and a robustness ranking that predicts downstream twin→real success rate (SR) with Spearman $\rho \geq 0.6$. Downstream SR comes from navigation policies trained on each frozen encoder in the twins and tested in tier A. Decision rules are those of §7.

Stage II – policy zoo and transfer forecasting (RQ2, H2; sem. 4–5).

- *Policy zoo:* ≥ 200 point-goal and image-goal navigation policies trained in the twins by reinforcement learning (PPO / DD-PPO) and imitation of a privileged planner. They vary scene, capture budget, encoder, architecture, training data and seed. Each policy is evaluated in its twin and in tier A, which gives the label: the twin→real SR gap. Small policies and shared frozen encoders keep the zoo within academic compute (a pilot in sem. 3 fixes the final size).
- *Predictors:* (i) a graph neural network over per-layer hidden-state statistics on a fixed probe set of twin frames and, where available, unlabelled real frames of the capture (no real rollouts; the



student's prior method for forecasting from hidden states); (ii) weight-space metanetworks [26, 27]. *Baselines*: twin SR alone, image fidelity (PSNR/LPIPS), and simulator-level predictivity: real SR predicted from twin SR by the sim-to-real mapping fitted on training scenes [10]. Splits are by held-out scene. Ablations keep fidelity fixed and vary the policy, so that the predictor cannot rely only on twin quality.

- *Failure monitors*: hidden-state failure scores (in the style of [29, 30]) with thresholds set by split conformal prediction [28] at 90% coverage in the twin; coverage is measured on tier A closed-loop rollouts (tier C reports agreement).
- *Success criterion (H2)*: $\geq 20\%$ lower MAE than the best baseline (lower 95% bound of the reduction > 0), and real coverage $\geq 85\%$ ($\epsilon = 5$ pp).

Stage III – budget-allocation experiments (RQ3, H3; sem. 5–6).

- *Levers*: (a) weighting of twin training data by representation distance to a small real set; (b) capture selection, i.e. which scenes or regions to (re)capture, guided by the localized gap from Stage I; (c) rollout selection, i.e. which real rollouts to collect, guided by the Stage II forecasts and their uncertainty. Real SR is estimated from few rollouts with prediction-powered inference [33, 34].
- *Baselines*: uniform and random allocation of the same budget. Analysis: attribution of real successes and failures to twin data [31, 32].
- *Success criterion (H3)*: budget–performance curves (real data = capture minutes + real rollouts in one operator-time cost) fitted per method; guided allocation reaches the target SR τ with ≤ 0.7 of the uniform budget (upper 95% bound of the ratio < 1).

Stage IV – cross-task and cross-simulator generalization (RQ4, H4; sem. 6).

- *Manipulation set*: public manipulation policies evaluated in the SIMPLER [12] replicas of real set-ups, with their published paired sim/real results as labels (tier B; ≥ 12 policy–task pairs). These policies are trained on real data, so the test asks whether performance in a simulated replica holds in reality.
- *Method*: the navigation-trained hidden-state predictor, which does not depend on the architecture, is applied without retraining. *Comparators*: SRCC of a generic simulator [10] and a frozen learned world model used as a policy evaluator (in the style of [35]); world models are not trained.
- *Success criterion (H4)*: Spearman $\rho \geq 0.5$ on manipulation, and a higher rank correlation with real outcomes than both comparators (95% CI of each paired difference excludes 0).

Tier C validation. Two robot campaigns (sem. 5 and 7) check that tier-A conclusions carry over: 8 zoo policies $\times$ 60 episodes (2 environments $\times$ 10 start–goal pairs $\times$ 3 trials), about 16 robot-hours per campaign at ~ 2 min per episode. Tier C is reported as agreement and is never used to tune thresholds.

Statistics and reproducibility. Tests are one-sided ($\alpha = 0.05$), bootstrapped over scenes and Holm-corrected within each hypothesis, as in §7. The protocol is pre-registered in the project repository before each stage, and changes are logged with reasons. Code, the paired-frame benchmark and the policy zoo are released where the dataset licences permit.

10. Streszczenie popularnonaukowe / Abstract for general public

Streszczenie popularnonaukowe należy przygotować w dwóch językach: polskim i angielskim. Wersje językowe muszą być tożsame. Opis powinien być napisany językiem popularnonaukowym i zawierać cel rozprawy, opis planowanych badań naukowych oraz powodów, dla których podjęta została ta tematyka badawcza, a także najważniejsze spodziewane efekty (maks. 1 strona, czcionka 11 pt., interlinia 1). / The description for general public must be in Polish and in English. The language versions must be identical. The description should be written for general public and should include the doctoral dissertation goal, description of research, reasons for attempting a particular research topic and substantial results expected (at most 1 page, font 11, line spacing 1).



Streszczenie popularnonaukowe

Roboty uczą się dziś samodzielnego działania, na przykład poruszania się po budynku lub chwytania przedmiotów, metodą prób i błędów. Potrzebują do tego bardzo wielu prób, więc zwykle ćwiczą w symulacji komputerowej. Coraz częściej jest nią cyfrowy bliźniak: wierna, fotorealistyczna kopia prawdziwego miejsca, którą metody sztucznej inteligencji potrafią odtworzyć z krótkiego filmu nagranego telefonem. Kopia nigdy nie jest jednak idealna. Bywa rozmyta, ma drobne zniekształcenia i inaczej oddaje światło. Robot, który świetnie radzi sobie w kopii, może więc zawieść w prawdziwym świecie. Dziś można to sprawdzić tylko, wypuszczając go do prawdziwego budynku, a to jest powolne i kosztowne.

Celem rozprawy jest sprawdzenie, czy da się to przewidzieć wcześniej, zaglądając „do głowy” robota. Model sztucznej inteligencji, który steruje robotem, przetwarza każdy obraz na wewnętrzny opis tego, co widzi. Zbadamy, czy obraz z kopii i obraz z rzeczywistości tego samego miejsca są przez model opisywane podobnie, w której części modelu pojawiają się różnice i jak maleją, gdy nagranie jest dłuższe. Następnie nauczymy osobny model przewidywać na podstawie tych wewnętrznych opisów, jak dobrze robot poradzi sobie w rzeczywistości i kiedy może popełnić błąd, zanim zostanie tam wysłany. Na koniec sprawdzimy, czy takie prognozy pomagają mądrze wydać niewielki zasób prawdziwych danych: co warto nagrać i które próby w rzeczywistości przeprowadzić. Głównym zadaniem testowym będzie nawigacja robota mobilnego, a sprawdzianem uniwersalności chwytanie przedmiotów ramieniem robota. Badania będą prowadzone przede wszystkim na publicznie dostępnych nagraniach i trójwymiarowych skanach prawdziwych budynków, a wybrane wyniki zostaną potwierdzone na prawdziwym robocie.

Spodziewanym efektem są metody, które pozwolą ocenić robota wyuczonego w cyfrowej kopii, zanim trafi do prawdziwego świata, oraz zmniejszyć liczbę kosztownych prób w rzeczywistości. Może to przyspieszyć i potanić wdrażanie robotów w magazynach, szpitalach, biurach i fabrykach. Opracowane oprogramowanie, dane i wyuczone modele zostaną udostępnione publicznie.

Abstract for general public

Robots today learn to act on their own, for example to move around a building or to grasp objects, by trial and error. They need a very large number of trials, so they usually practise in a computer simulation. Increasingly, this is a digital twin: a faithful, photorealistic copy of a real place that artificial intelligence methods can reconstruct from a short video recorded with a phone. The copy, however, is never perfect. It can be blurry, have small distortions and render light differently. A robot that does very well in the copy may therefore fail in the real world. Today this can only be checked by sending it into a real building, which is slow and costly.

The goal of this dissertation is to find out whether this can be predicted earlier by looking "inside the head" of the robot. The artificial intelligence model that controls the robot turns every image into an internal description of what it sees. We will study whether an image from the copy and an image from reality of the same place are described similarly by the model, in which part of the model differences appear, and how they shrink when the recording is longer. Next, we will teach a separate model to predict from these internal descriptions how well the robot will cope in reality and when it may make a mistake, before it is sent there. Finally, we will check whether such forecasts help to spend a small amount of real data wisely: what is worth recording and which trials in reality to carry out. Mobile-robot navigation will be the main test task, and grasping objects with a robot arm will test how universal the approach is. The research will be carried out mainly on publicly available recordings and three-dimensional scans of real buildings, and selected results will be confirmed on a real robot.

The expected result is methods that make it possible to assess a robot trained in a digital copy before it reaches the real world, and to reduce the number of costly trials in reality. This may make deploying robots in warehouses, hospitals, offices and factories faster and cheaper. The software, data and trained models developed will be made publicly available.



11. Termin oddania do druku artykułu naukowego lub monografii / Date of submission for publication of at least one scientific article or monography

Termin wysłania do recenzji artykułu naukowego lub monografii naukowej zgodnie z wymaganiami art. 186 ust. 1 pkt 3 Ustawy stawianymi osobie ubiegającej się stopień doktora. Termin dotyczy oddania do druku co najmniej jednego artykułu naukowego opublikowanego w czasopiśmie naukowym lub w recenzowanych materiałach z konferencji międzynarodowej w odpowiedniej dyscyplinie naukowej, które w roku opublikowania artykułu w ostatecznej formie były ujęte w wykazie („lista ministerialna”) sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. b) lub 1 monografii naukowej wydanej przez wydawnictwo, które w roku opublikowania monografii w ostatecznej formie było ujęte w wykazie sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. a). / A date of submission for publication of at least one scientific article published in a scientific journal or in the peer-reviewed materials of an international conference in the relevant scientific discipline, which in the year of publication of the article in its final form were included in the list drawn up in accordance with the regulations issued under the Act (Art. 267 sec. 2 point 2(b) or 1 scientific monograph published by a publishing house, which in the year of publication of the monograph in its final form was included in the list drawn up in accordance with the regulations issued pursuant to the Act (Art. 267 sec. 2 point 2(a), referred to in Art. 186 sec. 1 point 3 of the Act.

May 2027 / maj 2027

Paper **P1** submitted for review to the **Conference on Neural Information Processing Systems (NeurIPS 2027)**, main track or Datasets & Benchmarks track. NeurIPS is peer-reviewed international conference proceedings worth **200 points** on the ministerial list (conference Lp. 87), assigned to the discipline *informatyka techniczna i telekomunikacja*. P1 reports Stage I (RQ1): a benchmark of paired real and twin-rendered frames, the localization of the twin-vs-real gap inside frozen encoders, and a first test of whether encoder robustness predicts the transfer of twin-trained policies (task T4.2). If it is rejected, the revised paper goes to **CVPR 2028** (expected deadline November 2027, also 200 points), and then to the next 200-point conference (§12).

12. Inne / Other comments

Dodatkowe wyjaśnienia, komentarze, podsumowanie związane z indywidualnym planem badawczym (maks. 1 strona, czcionka 11 pt., interlinia 1). / Additional explanations, comments, summary related to the individual research plan (at most 1 page, font 11, line spacing 1).

Planned outputs by year. All papers target 200-point conferences assigned to the discipline (list of 5.01.2024; re-checked against the new list).

Year	Papers (planned submission, 200 points each)	Data and code	Grants	Mobility, events
1 (2025/26)	—	—	—	—
2 (2026/27)	P1 (RQ1, paired-frame benchmark and gap localization): NeurIPS 2027, May 2027 (§11)	Benchmark v1	SzD Minigrant; NCN PRELUDIUM; NAWA Bekker	Summer school poster



3 (2027/28)	P2 (RQ2, forecasting transfer and failure): CVPR or ICLR 2028; P3 (RQ3–RQ4, budget allocation, manipulation): NeurIPS 2028	Policy zoo	—	Mid-term; 3-month visit (sem. 6)
4 (2028/29)	Optional P4, consolidated study: ICLR 2029 or CVPR 2029	Full release	—	Dissertation

P1–P3 are the thematically linked core of the dissertation; P4 is optional.

Foreign research visit (candidate hosts, not yet contacted). (1) Z. Kira's group, Georgia Tech: EmbodiedSplat [4]. (2) Multi-robot Systems Lab, Stanford University (M. Schwager): Gaussian-splatting scene models and navigation. A European host (Erasmus+ route) from the weight-space learning community (authors of [25–27]) and a foreign co-author for P2 are sought with the supervisor in sem. 4.

Collaboration at PWr. Real-robot validation is planned with the *Denali* Autonomous Robots Laboratory (K29, W12N; Pioneer 3-DX, Jaguar 4x4, ROS 2); a written agreement is task T3.2 (Nov 2026).

Risks and mitigation (semester affected).

- *Policy zoo too small* (4–5): small policies, shared frozen encoders, imitation of a privileged planner; the sem. 3 pilot fixes the size; WCSS and PLGrid grants.
- *Twin–real pairs unavailable* (3): licence or quality limits on the chosen dataset. Other public datasets with real captures and reference scans, and own captures at PWr.
- *Scooping* (3–6): failure detection and policy interpretability move fast. Early preprints, benchmark release, monthly literature monitoring; the zoo and weight-space angle differentiate the work.
- *Predictor learns only twin fidelity* (5): ablations with fixed fidelity; a negative result still answers RQ2.
- *Proxy reality differs from reality* (5–7): tier C campaigns check agreement; they can move by a semester.
- *Rejection at a top venue* (4–7): every paper has a resubmission path inside the 200-point set, 2–4 months later (e.g. NeurIPS → CVPR → ICML → ECCV/ICCV), so the degree requirement does not depend on one decision.

Prior work (background only). A pre-PhD NLP paper (ACL 2025 SRW) forecasts from hidden states with a graph neural network over layers; Stage II adapts the method, but the paper is not part of the dissertation.

Ethics and data. Own captures anonymize personal data (GDPR); public datasets are used under their licences.

13. Określenie planowanej formy współpracy z promotorem / Planned form of cooperation with the supervisor

Krótki opis planowanej formy współpracy z promotorem, np. częstotliwość spotkań, forma spotkań i ich zakres, sposób komunikacji. / Brief description of the planned form of cooperation with the supervisor, e.g., frequency of meetings, form of meetings and their scope, method of communication.

- **Weekly meeting** (~1 h, in person at K46 or online via MS Teams) to discuss progress, results and next steps.
- **Monthly written progress summary** (completed tasks, results, problems, plan). It lists every IPB task from §3 with its % **completion**, in the same form as the task table of the mid-term self-report.



- **Semester review** against the IPB schedule (§3), before each annual report and the mid-term evaluation. Deviations from the plan and their reasons are recorded and agreed with the supervisor.
- **Supervisor co-signs** the annual reports and the mid-term self-report (autoreferat).
- **Paper-writing sprints** before each submission deadline, with joint editing in Overleaf. The supervisor co-authors and reviews the papers before submission.
- **Ongoing communication** via e-mail / Teams chat. The code and experiment logs are in a shared Git repository that the supervisor can access.

14. Opinia promotora pomocniczego / Opinion of assistant supervisor

Promotor pomocniczy (jeżeli jest wyznaczony) powinien zaopiniować IPB. / Assistant supervisor (if appointed) should provide an opinion of the IRP.

nie dotyczy / not applicable

.....
Podpis promotora pomocniczego /
Signature of assistant supervisor

15. Podpisy doktoranta i promotora/promotorów / Signatures of doctoral student and supervisor(s)

Wybierz datę/Select date

.....
Data / Date

.....
Podpis doktoranta / Signature of doctoral student

.....
Podpis promotora / Signature of supervisor

.....
Podpis drugiego promotora / Signature of second supervisor